

Automation in Inventory Redistribution within a Hybrid Supply Chain with Reverse Logistics: A Heuristic Approach Based on Business Rules

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Abstract—Efficient management of empty glass bottle collection from distribution facilities to central processing units in reverse logistics reduces operational costs and supports environmental sustainability. This study examines a practical business case within a hybrid closed-loop supply chain, where new products replace losses, breakages, or wear to maintain a steady supply across processing centers. This ensures a continuous flow of materials essential for preventing production disruptions and avoiding bottlenecks when integrating new products. To address these challenges, the study focuses on scheduling material collection with two primary objectives: preventing stock shortages that could disrupt production due to insufficient bottles and avoiding congestion at facilities where new products are introduced. The proposed approach integrates real-time analysis of manufacturing data, return flows, center saturation levels, and inter-center transportation with heuristic-based decision-making for transport selection between centers. The computational experiments demonstrate the effectiveness of the automation process, ensuring timely material movement and improving overall supply chain efficiency.

Index Terms—Reverse logistics, hybrid supply chains, and heuristics.

I. INTRODUCTION

Glass containers are widely used in the beverage industry for their ability to preserve flavor and product quality. Historically, nearly all carbonated beverages were packaged in glass bottles with crown stoppers [1]. However, single-use glass has the highest carbon footprint across all beverage types and regions [2], contributing significantly to global warming, ozone depletion, acidification, fossil resource depletion, water consumption, and human carcinogenic toxicity in life cycle assessments [3]. In contrast, recyclable and refillable glass bottles offer clear ecological advantages by reducing resource waste. While they eliminate the recurring cost of purchasing

new containers, the required investment in washing facilities, bottle pools, and logistics structures can be substantial [4]. Despite these costs, the combined environmental and economic benefits make refillable glass bottles a logical choice for companies packaging their products in glass.

In the circular economy, recycling and reusing raw materials or products have become essential components of business operations. Reverse logistics plays a central role in managing the flow of materials after they have served their initial purpose, facilitating their collection, transportation, and redistribution for reuse or recycling [5]. It is commonly defined as the process of planning, implementing, and controlling the efficient return and recovery of products and materials, reducing environmental impact while improving resource efficiency [6]. This approach enhances resource utilization, minimizes waste, and lowers carbon emissions, ensuring that recovered materials are effectively reintegrated into the supply chain [7]. Recycling within reverse logistics is particularly relevant in industries with high material recovery potential. The strategic collection and redistribution of used products contribute significantly to environmental and economic sustainability [8]. However, effective logistics management requires integrating warehousing, production planning, and transportation scheduling to adapt to fluctuating demand. This improves the accuracy, reliability, and responsiveness of the supply chain [9], [10]. Decision-making in this context extends beyond traditional transportation problems that focus solely on balancing supply and demand. The coordination of intra-company logistics must align production planning with inventory management, resource allocation, and transportation optimization to achieve cost reductions and lower carbon emissions while meeting production requirements. Specifically, the reuse of empty glass bottles in the beverage industry plays a significant role in sustaining operations. Transportation planning directly

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influences both efficiency and business continuity, as factors such as storage capacity, safety stock requirements, vehicle availability, and routing restrictions – combined with multiple operational objectives – add complexity to daily decision-making.

This study examines the transportation planning problem faced by a beverage company, focusing on the efficient collection and distribution of empty glass bottles to production centers in need. This problem extends the traditional transportation or shipping problems into a realistic network optimization challenge, where empty glass bottles must be efficiently transported between multiple nodes, including warehouses and factories. Several operational constraints must be addressed to ensure feasibility. Key requirements include maintaining adequate stock levels without overloading warehouses, adhering to a three-week production schedule based on forecasted demand, and minimizing both transportation costs and environmental impact. Additional complexity arises from the hybrid nature of the supply chain, where returned bottles are supplemented with new glass bottles to replace those that are broken, unreturned, or lost [11]. Addressing these challenges requires consideration of factors such as return rates, production variations, alert prioritization, and specific constraints of the beverage production line, including the need to maintain a required proportion of old and new glass bottles.

In practice, these complex problems are often addressed using traditional methods such as manual planning, which is inefficient, costly, and prone to decision errors due to incomplete or overlooked information. [12] highlights the shortcomings of traditional, non-integrated production planning in the steel manufacturing industry, citing issues such as isolated and inconsistent information processing, difficulties in data exchange, and delays in updating critical information. Similarly, [13] demonstrates the time-consuming and often unsatisfactory outcomes of manual production planning and scheduling in the pulp and paper industry. To overcome these challenges, this paper proposes an automated decision-making solution designed to manage operational complexities more effectively. A customized algorithm dynamically generates optimized transportation plans by incorporating real-time data inputs, including production schedules, inventory levels, and return forecasts, along with additional factors such as warehouse saturation levels, existing transportation routes, and vehicle availability. Automating routine bottle transportation decisions allows for a more responsive and efficient system, ensuring production continuity while addressing out-of-stock warnings across all factories and meeting regulatory requirements.

The key contributions of this paper are as follows: (i) it formulates the routine operational problem as a priority-driven, multi-stage heuristic redeployment model, incorporating essential constraints such as demand fulfillment, warehouse and truck capacities, movement priorities, buffer management, and debt transfer; (ii) it proposes a solution approach that integrates rule-based and heuristic algorithms with real-time warehouse data to enable fast and efficient decision-making;

and (iii) it validates the proposed solution using real operational data from a beverage company, demonstrating how automated scheduling enhances accuracy in meeting management requirements and improves overall operational efficiency. This study provides a practical framework for addressing transportation challenges in the beverage industry and other sectors. The proposed solution not only automates routine operational decisions but also establishes a foundation for scalable, data-driven logistics management that supports future growth and operational flexibility. Furthermore, the model is applicable to industries dealing with multi-warehouse logistics and inventory balancing, contributing to the broader field of supply chain optimization.

The remainder of the paper is structured as follows. Section II reviews related work on similar problems. Section III defines the problem statement. Section IV details the automated decision-making algorithm and presents its implementation. Section V discusses the computational experiments and obtained results. Section VI provides managerial insights from the automation process. Finally, Section VII synthesizes key findings, discusses methodological limitations, and outlines directions for future research.

II. RELATED WORK

The transportation problem is a classical optimization problem in linear programming that focuses on minimizing shipping costs while transferring commodities from multiple sources to multiple destinations [14]. The foundational model was introduced by [15] and later solved using the simplex method [16]. Over time, researchers have developed various operations research and management science (OR/MS) techniques to tackle transportation problems. These problems are typically addressed through three main steps: mathematical formulation, obtaining an initial feasible solution, and optimizing that solution. Methods for determining an initial feasible solution include the northwest corner rule [17], Vogel's approximation [18], and the allocation table method [19]. In addition to exact methods, metaheuristic algorithms such as genetic algorithms [20], ant colony optimization [21], particle swarm optimization [22], and tabu search [23] have been widely applied to improve efficiency.

With increasing complexity in real-world transportation problems, traditional mathematical models have struggled to fully capture practical constraints. Expanding the problem by incorporating vehicle diversity and varying capacities has led to the development of solid transportation models [24]. While classical formulations assume fixed routes, alternative routing approaches have been explored by introducing additional dimensions. For instance, [25] extended the problem by integrating route choices based on origin, destination, and vehicle type, transforming it into a four-dimensional transportation problem. Green transport scheduling has also been examined with added constraints such as pick-up times and transport modes [26]. Cross-docking networks have further refined the problem by optimizing the movement of goods between suppliers and retailers, with integer programming

and ant colony optimization effectively reducing costs and improving delivery efficiency [21].

Sustainability has become a key consideration in transportation optimization due to increasing environmental concerns and regulatory pressures. While traditional models primarily focused on cost and time efficiency, more recent studies have incorporated environmental objectives such as minimizing emissions, reducing fuel consumption, and promoting renewable energy use in logistics. The transshipment problem for electric vehicles has been investigated, with a multi-objective battery-swapping model demonstrating reductions in both transportation costs and travel time [27]. Carbon taxes have also been integrated into optimization models to internalize environmental costs and encourage sustainable routing and vehicle selection [26]. By incorporating carbon emissions into a multi-objective green transportation model, [28] accounted for factors such as distance, vehicle type, and path conditions when determining optimal solutions.

Despite the increasing number of studies integrating inventory and transportation decisions, practical implementation remains a challenge. Many models oversimplify real-world conditions, resulting in a gap between theoretical approaches and their applicability [29]. This study aims to bridge that gap by developing a practical approach that effectively combines inventory management and transportation planning within a real-world logistics system. The proposed solution is designed to address the specific needs of the company under study while providing a framework applicable to similar logistics challenges.

III. PROBLEM STATEMENT

The paper addresses the transportation planning problem faced by a beverage company, focusing on the efficient collection and distribution of empty glass bottles to production centers. This problem presents two key challenges that must be addressed to ensure efficient stock and logistics management. First, stock shortages must be prevented across production centers to ensure the availability of bottles required for manufacturing over the next four weeks. Second, the central center, which serves as the entry point for new product inventory, must maintain a balance to prevent stored stock from exceeding 100 truckloads at the end of the distribution process. Avoiding saturation minimizes excessive storage costs and operational constraints. To address these challenges, the following objectives are established:

- Ensure the transfer of trucks from centers with sufficient stock to those requiring specific bottle models, based on their production demand.
- Optimize the transportation of glass bottles, prioritizing the use of fully loaded double-height trucks to reduce transportation costs and carbon footprint.
- Maintain a safety stock for each product at all centers, considering production variability and the percentage of returned glass.

- Limit stock levels at the central center, ensuring an efficient redistribution of excess stock to other centers to prevent saturation and operational issues.

The case study involves nine centers across different regions, categorized into four types: producing centers, non-producing centers, buffer centers, and the central center. Given the wide range of products and bottle formats, not all centers manufacture every product. Bottle format variations impact truck load capacities, necessitating compliance with legal weight restrictions. Non-producing centers receive daily glass bottle returns from customers, who send them to the nearest center. Consequently, these centers may store returned products they do not produce. To free up storage space, non-producing centers must transfer such products to the appropriate producing centers. Additionally, the company differentiates between major and minor products. Major products, in higher demand and produced frequently, are prioritized in transportation planning. Minor products require detailed scheduling to prevent bottlenecks and inefficiencies. Buffer centers serve as strategic storage hubs, managing excess stock. The central center plays a dual role: distributing returned bottles collected within its region and acting as the sole distributor of new bottles to other centers. Predefined routes exist between centers, with a maximum number of trucks allowed per route. Any logistics solution must adhere to these constraints. Figure 1 illustrates the problem, depicting the central center as the hub with other centers surrounding it. Some centers incorporate buffers for stock management. Dashed lines represent intercompany routes accessible only to conventional trucks, while solid lines indicate routes from the central center to factories for new bottle distribution, where both conventional and double trucks are permitted.

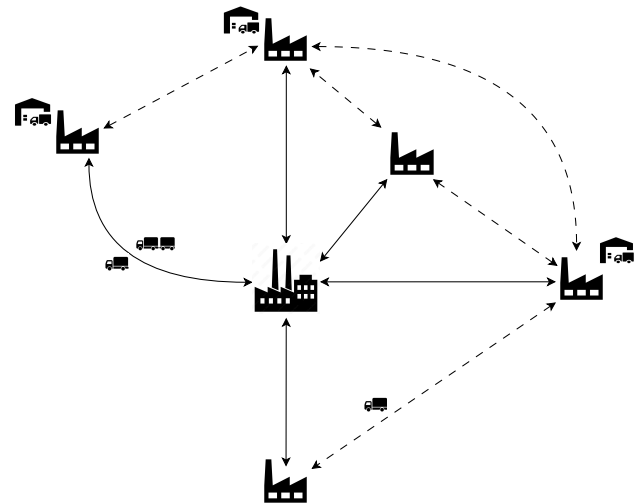


Fig. 1. Simplified representation of the glass transportation problem.

The company uses different types of trucks for transportation operations, each with a specific load capacity. To standardize logistics planning, the equivalent truckload (CCE)

is used as the unit of measurement for product transfers. One CCE corresponds to a fully loaded conventional truck carrying 33 pallets, with the exact number of boxes and bottles varying by product. This standardization facilitates transportation planning and optimization, ensuring that logistics decisions are expressed in equivalent truckloads. Therefore, the capacity of the different available truck types is defined as follows:

- Conventional truck: 1 CCE
- Double-height truck: 2 CCE
- HVO truck: 1 CCE
- Double-cabin truck: 2 CCE

IV. DECISION-MAKING AUTOMATION PROCESS

The solution approach focuses on enhancing expert-based manual planning by integrating automation to improve sustainability and operational efficiency. Automated decision-making ensures more accurate and timely responses, contributing to a streamlined logistics operation. A flexible system is essential, providing alternative feasible solutions based on priorities and constraints while allowing real-time adjustments without compromising performance. In addition, maintaining continuity in the decision-making process is essential. Assigning resources to resolve stock shortages affects future availability, as both the allocated product and the transportation used become unavailable for subsequent assignments. To account for these dependencies, the system continuously updates datasets (DB), reflecting real-time stock levels and transportation availability to support informed decision-making.

Figure 2 presents an overview of the decision-making automation process, illustrating the relationship between the two challenges. To address stock shortages in producing centers, Heuristic 1 is applied, which requires as input all relevant datasets (stock levels, production, safety stock, returns, transportation routes, etc.), as well as the warning products (WP). These are products in centers where a stock shortage is anticipated. This heuristic generates an initial solution by determining the movement of trucks carrying WP from an origin center to a destination center in need of stock. Additionally, it updates all utilized datasets to reflect the assigned resources. These updated datasets, along with the planned transport trips from the central center, serve as inputs for Heuristic 2. In cases where the origin center selected in Heuristic 1 is the central center, these movements must be incorporated into the predefined transportation routes to avoid duplication. This step ensures that resources are not allocated twice in the optimization process. The final solution resolves stock shortages while integrating product movements efficiently. Movements originating from the central center are optimized, ensuring that all designated trips are correctly assigned and resources are utilized effectively.

The remainder of the section presents the two heuristics used to obtain solutions to the identified challenges. In these solutions, the movement of a specific product is always defined from an origin center to a destination center, along with the type of transportation to be used. The transport type is assigned as follows:

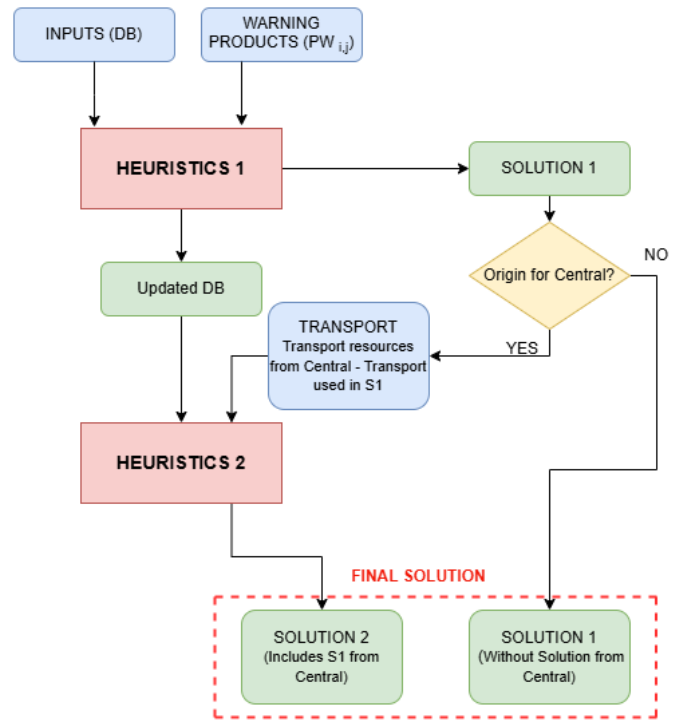


Fig. 2. Flowchart of the general process.

- In stock shortage problems where the origin center is not the central center, all trips are carried out using conventional trucks, taking advantage of routes where trucks deliver products and return with empty glass bottles.
- In the case of trips departing from the central center, predefined routes exist with a specific type of transportation, allowing for optimized truck utilization and ensuring efficient distribution.

A. Heuristic 1: Avoiding stockouts

The proposed heuristic prevents production interruptions and ensures demand fulfillment through efficient stock distribution. The heuristic begins with detecting, identifying, and classifying stock alerts into two priority-based lists. Red alerts indicate an immediate stock-out for major products or an anticipated shortage for minor products within three weeks, while orange alerts signal stock levels falling below safety thresholds or a minor product shortage in the fourth week. Red alerts are assigned to Priority List 1, as they require urgent resolution, whereas orange alerts are placed in Priority List 2. Once alerts are classified, predefined rules are applied to address them, prioritizing actions for the most critical shortages first.

The logic followed is based on a sequence of steps designed to address warnings through predefined movements. Initially, products are redistributed from non-producing centers to facilities with active warnings. Each center collects empty bottles from its designated area, and if it does not manufacture a specific product, the necessary stock is redirected to the appropriate facility. Even in the absence of active

warnings, this redistribution remains essential for optimizing center capacity. If stock transfers from non-producing centers prove insufficient and alerts persist, additional measures are implemented to resolve priority lists and address critical red warnings. The next step involves supplying new empty glass bottles from the investment facility, located in the central center. However, since each producing center receives a fixed allocation of new bottles, this action alone may not fully resolve the warning. If necessary, bottles designated for other centers can be reallocated to the affected facility, though this increases intercompany debt. When no additional bottles are available, truck transfers between facilities serve as a final measure to mitigate intercompany debt. After executing all steps, the results of heuristic 1 are systematically generated and structured for decision-makers. These results include recorded movements, applied solutions ranked by priority (List 1 or List 2), and a summary of actions taken, such as stock reallocations, bottle distributions, or debt adjustments. If any alerts remain unresolved, the output functions as a notification system, signaling the need for manual intervention or further analysis.

B. Heuristic 2: Optimization of Central Center Logistics

The proposed heuristic optimizes the redistribution of empty glass bottles from the central center to designated producing centers. Acting as both an entry point for new bottles and a collection hub for recycled ones, the central center must efficiently manage stock flow, as it lacks production capabilities. Predefined transportation routes facilitate this process, preventing unnecessary stock accumulation, reducing storage costs, and maintaining space for incoming bottles. In addition, it considers vehicle suitability for transport, center saturation levels, stock availability at the central center, and the priority assigned to each bottle type at individual producing centers. Distribution follows a structured sequence, prioritizing centers with the least available transportation capacity to ensure equitable resource allocation. As the process advances to centers with greater capacity, the system dynamically adjusts, optimizing weekly movements and maintaining a balanced, cost-effective supply chain.

The sequence of the distribution process from the central depot follows a defined logic that guides each stage of the procedure. Initially, the heuristic first checks whether the distribution destination center is operating above its maximum saturation level. If the threshold is exceeded, empty bottle distribution is halted to prevent overstocking. Otherwise, the heuristic evaluates truck availability at the central center for distributing both new and recycled products. The goal is to maintain fewer than 100 trucks in stock at the central center by the end of the process. If the initial stock is already below this threshold, stock movements are adjusted based on priorities, reducing trips when necessary. If stock levels exceed 150 trucks, additional transportation routes between distribution destination centers are identified to maximize utilization. At this stage, transportation capacity for stock distribution is determined to ensure an efficient and balanced allocation of

resources. Next, priority lists for empty bottles are created at each distribution destination center. These priorities depend on production planning, inter-center packaging debt, and warning products—items projected to fall below safety stock levels due to future production demands. The priority list dictates how available stock at the central center is allocated to each destination efficiently. Two separate lists are managed: one for new bottles and another for recycled bottles. Since production quotas limit the use of new empty bottles, this distinction ensures distribution aligns with production requirements while adhering to predefined constraints. Once priority lists and transportation resources are defined, the distribution of new empty bottles begins. When this step is completed and no stock of new bottles remains, the system updates truck availability. Any unused trucks are then allocated for distributing recycled bottles according to the priority list. In cases where all transportation resources are used for new bottles, the heuristic solution concludes with their allocation. However, if trucks remain available, the solution consists of a combined distribution of new and recycled bottles.

V. COMPUTATIONAL EXPERIMENTS

The decision-making automation process incorporating both heuristics was implemented in Python [30]. This programming language was selected for its flexibility and efficiency in handling large datasets and automating decision-making processes. Several external libraries, including Pandas and NumPy, supported data processing and the analysis of structured datasets. Once the code was developed, results were validated weekly in collaboration with the company to compare algorithm-generated solutions with actual decisions.

Figure 3 presents a set of heat maps representing the distribution of products and the utilization of different truck types across four producing centers (W1, W2, W3, and W4) for a specific week. The dataset includes various empty glass bottles, both new (PB_n) and recycled (PA, PB, PC, PD, PE), as well as transport resources used in each center (DA, Conv, and HVO), randomly distributed. The total truck stock in the central hub ranges between 100 and 150 vehicles, ensuring efficient logistics through an optimized transport distribution. Three different heat maps illustrate and compare truck allocation approaches:

- 1) Manual planning: The first heat map displays truck allocation based on the company's operational logic and experience during planning meetings.
- 2) Algorithm solution: The second heat map shows the truck allocation proposed by the algorithm, which optimizes distribution while adhering to model constraints and rules.
- 3) Difference between both solutions: The third heat map highlights discrepancies between manual planning and algorithm-generated solutions. Positive and negative values indicate centers where significant differences occur, identifying areas requiring further logistical adjustments.

This approach enables systematic solution comparison and validation, providing insights into the extent of errors. In this

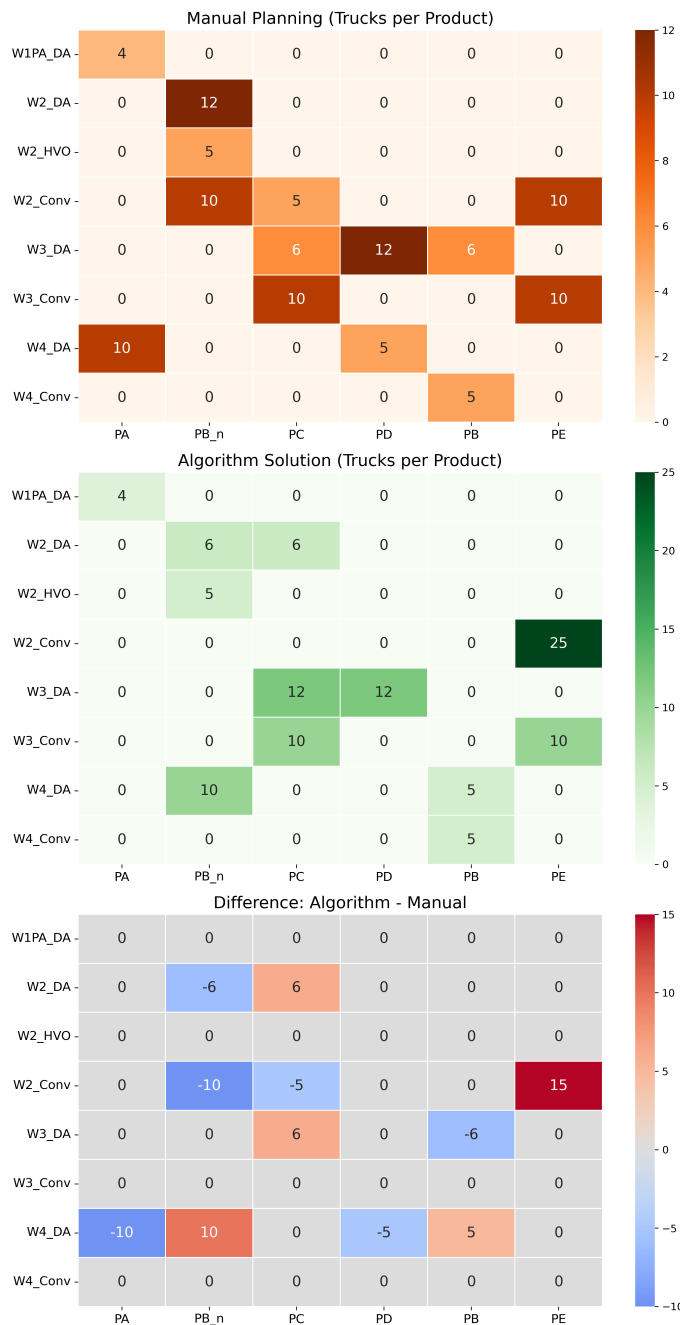


Fig. 3. Comparison of solutions taking into account product, center, and truck type in CCE.

example, the largest discrepancy is observed at center W2 in the allocation of conventional trucks, affecting both quantity and product type. This suggests a notable deviation between manual planning and algorithm-generated solutions. Differences in other centers typically involve product substitutions, meaning one product is dispatched in place of another.

Figure 4 presents a per-center comparison of the total CCE received for each product. This visualization contrasts CCEs allocated during operational planning with those dispatched based on the algorithm's recommendations. The graphs reveal

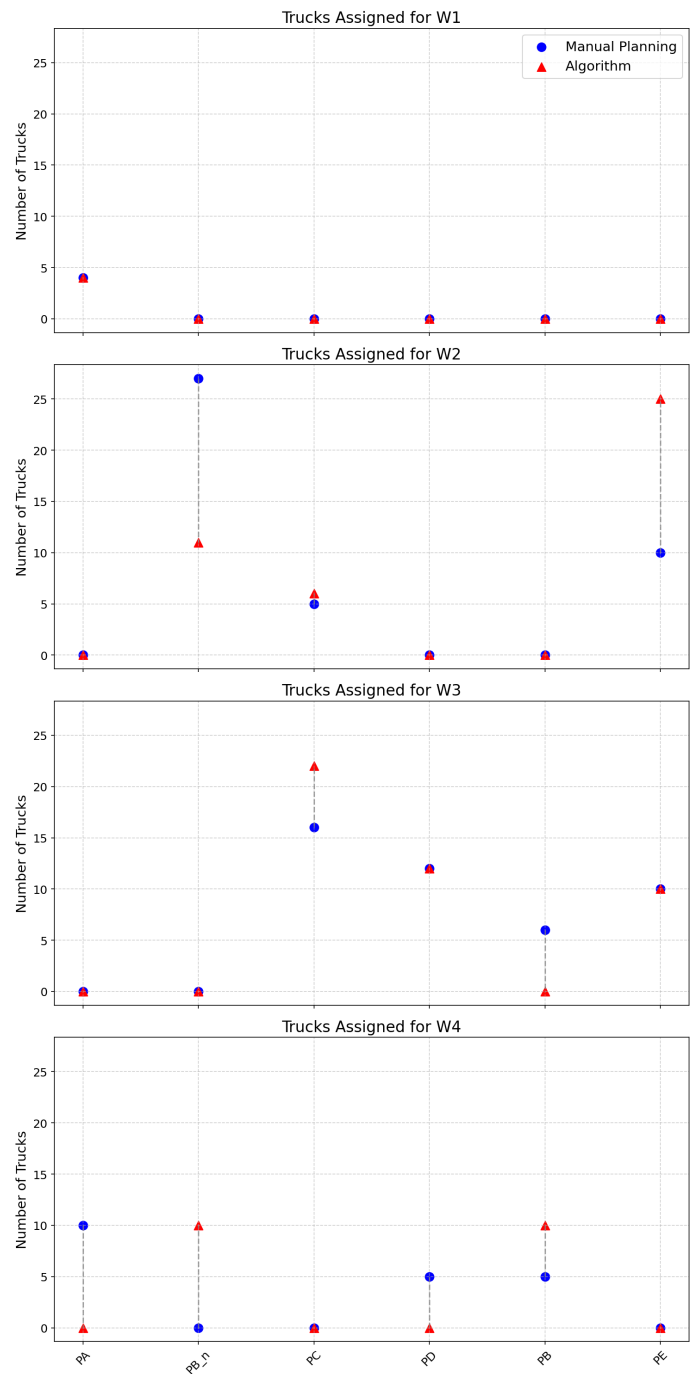


Fig. 4. Comparison of solutions between algorithm and reality taking into account product, center in CCE.

whether the algorithm dispatched more or fewer products than originally planned. The sequence of resource allocation is an important factor influencing distribution efficiency. At center W3, for instance, the algorithm prioritized distributing product PC over PB. As a result, when trucks were assigned for PC, fleet capacity was reached, preventing the dispatch of PB despite available stock. This analysis contributed to refining product prioritization lists, ensuring correct assign-

ment sequences. The algorithm minimizes truck overloading with a single product while preventing shortages of others by structuring the distribution order effectively.

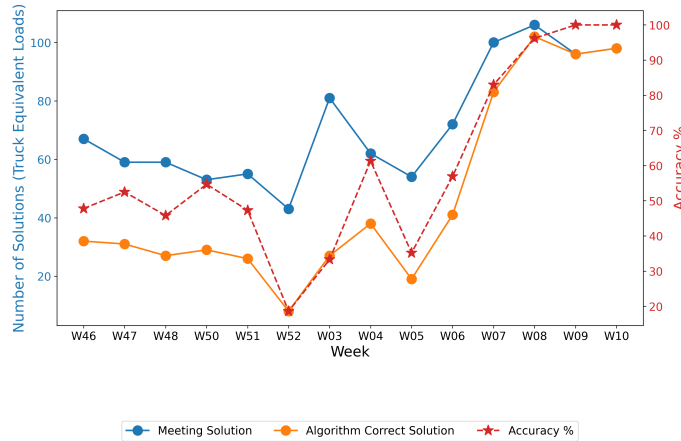


Fig. 5. Evolution of algorithm tuning in CCE.

Figure 5 illustrates this evolution, comparing total CCE values from the company's weekly planning with those generated by the algorithm, considering only correctly allocated units (i.e., excluding CCEs for different products or those exceeding required quantities). The accuracy rate of the algorithm's weekly predictions is also depicted. As adjustments progressed, algorithm-generated solutions increasingly aligned with operational planning. During the final testing weeks, the model replicated the company's planning with high precision, demonstrating appropriate constraint parameterization and effective alignment between algorithmic optimization and decision-making processes.

Throughout this research, progressive adjustments have been made to the constraints and logic used by the company to enhance the alignment between algorithm-generated solutions and operational decisions. Over more than 20 weeks of execution, Heuristic 1 consistently provided stable and precise solutions. However, stock distribution from the central center to the producing centers showed inconsistencies across different weeks. While some solutions closely matched company decisions, others varied due to production changes, fluctuations in bottle returns, and product seasonality. To improve model precision and adaptability, constraints incorporating seasonal variations were progressively introduced, ensuring better alignment with the real dynamics of the logistics system. Continuous validation across different weeks revealed that, while the algorithmic solutions were correct based on defined logic, they did not always fully align with company decision-making. This led to refinements in constraints and limitations, particularly in determining maximum saturation levels for each center and defining the number of available trucks to allow transport adjustments. These improvements have enhanced the algorithm's ability to reflect real-world logistical conditions, optimizing decision-making and increasing process efficiency.

VI. MANAGERIAL INSIGHTS

This automated decision-making process has significant managerial implications for large manufacturing and distribution companies. In modern operations, where diverse product lines are managed across multiple facilities, decision-making complexity continues to grow. Fluctuating market demand, increasing product variety, and uneven resource distribution require companies to make efficient inventory management and logistics scheduling decisions in a dynamic environment. Despite these challenges, many companies still rely on lengthy meetings where individual departments develop production plans separately. This manual approach often results in redundant information, delays, and difficulties in updating production conditions [12]. Cross-facility dependencies, real-time inventory alerts, and logistical constraints further complicate planning. As outlined in Section V, manual decision-making may overlook management priorities or address out-of-stock alerts inefficiently. Errors in this process can lead to oversaturation, understocking, and reduced operational efficiency, potentially causing serious production disruptions that weaken competitiveness and customer satisfaction. A heuristic-based automated decision-making algorithm enhances inventory scheduling and logistics operations by dynamically identifying and classifying inventory alerts. It integrates rule-based prioritization to evaluate and optimize decisions efficiently, offering a clear strategic advantage.

Key benefits of this automated decision-making process in inventory management and logistics scheduling include:

- 1) Ensuring production continuity: Tailored heuristics incorporate operational details to prioritize critical shortages. Red and orange alerts categorize product shortages by urgency, allowing the algorithm to mitigate risks to beverage production. Proactive monitoring of inventory levels helps prevent saturation at central facilities by reallocating resources to centers with production needs, ensuring uninterrupted operations.
- 2) Reducing waste and costs: The system dynamically balances stock levels across facilities, minimizing overstocking and storage costs while preventing resource waste. Existing facilities and resources are prioritized over external alternatives, such as new bottle production or additional shipping routes, leading to cost savings and reduced reliance on manual oversight.
- 3) Enhancing decision quality and reliability: Inventory forecasts and alarm hierarchy rules support logical and consistent allocation strategies, reducing human bias and errors. The system provides a comprehensive solution that accounts for safety stock levels, inventory saturation, and facility utilization while maintaining production schedules. Detailed action recommendations and clear reports on unresolved alerts enable decision-makers to respond efficiently.
- 4) Improving decision-making efficiency: The algorithm generates optimized solutions in under five seconds, allowing companies to react swiftly to dynamic in-

ventory alerts. Real-time adjustments to allocation and shipping strategies prevent decision delays that could disrupt production or create logistics bottlenecks. Faster responses to changes enhance order fulfillment speed and customer service, providing a competitive edge.

- 5) Supporting sustainability initiatives: Optimized allocation of recycled empty bottles maximizes resource utilization under the circular economy model. Prioritizing fully loaded double-height trucks reduces transportation frequency and fuel consumption, helping the company achieve sustainability goals.

Designed to meet the specific requirements of this real-world case, the heuristic approach enhances inventory management and logistics planning by aligning material allocation with facility and resource constraints while maintaining production schedules. It reduces costs, improves competitiveness, and supports long-term production planning for sustained efficiency in the supply chain network. While developed for a specific use case, this automated decision-making approach, based on rule-based heuristics and priority frameworks, can be adapted to other manufacturing companies facing similar logistical challenges. Its principles may also be applicable across industries such as retail, finance, and healthcare, offering broad potential for improving operational efficiency and strategic planning.

VII. CONCLUSIONS

This study developed an automated decision-making approach to optimize glass bottle transportation and inventory management in a multi-facility manufacturing environment. A heuristic-based model was designed to structure logistics planning efficiently while incorporating key operational constraints. The approach accounted for demand fluctuations, warehouse capacities, transportation priorities, and stock balancing across multiple locations. The methodology was tailored to align with real-world operational requirements, ensuring its practical applicability in a manufacturing setting.

The validation process demonstrated that the algorithm successfully replicated and improved the company's existing planning process. Over multiple weekly iterations, the model adapted to operational constraints, achieving greater accuracy and reliability in logistics planning. Key benefits included improved stock management by optimizing material allocation, reduced transportation costs through efficient truck utilization, and faster response times to stock alerts, with decisions generated in under 5 seconds. Additionally, the model's flexibility allowed real-time adjustments to production variability, return rates, and changing logistics conditions, preventing overstocking, understocking, and transportation inefficiencies. The implementation of heuristics proved to be a viable alternative to manual planning, reducing errors and enhancing consistency in decision-making.

Future research could explore integrating predictive modeling techniques to enhance forecasting accuracy and improve decision-making in dynamic supply chains. Expanding the framework to incorporate additional uncertainties, such as

demand volatility or disruptions in transportation networks, would further strengthen its robustness. While this study focused on a specific manufacturing case, the underlying methodology can be adapted to similar logistics challenges in retail, distribution, and broader supply chain environments.

VIII. MATCH & CONTRIBUTION

This contribution aligns with the theme of ICE IEEE 2025 on "AI-driven Industrial Transformation." The paper presents an automated, heuristic-based approach to optimize reverse logistics within a hybrid supply chain focused on the collection and redistribution of glass bottles. Through real-time analysis of production and transportation data, the approach enhances operational efficiency and supports sustainability.

The contribution is particularly relevant to the conference areas related to Process Optimization and Smart Production, Sustainable Engineering and Circular Manufacturing, and AI-enhanced Data-Driven Decision Making. It presents a real-world business case demonstrating how heuristic methods, supported by digital transformation tools, can automate complex logistics decisions, reduce transportation costs, and strengthen supply chain resilience. The model's implementation provides a practical, AI-adjacent solution that bridges operational constraints with strategic planning.

Additionally, the model's ability to adapt to changing conditions and its potential scalability to other sectors reinforce its relevance to broader topics such as Smart Manufacturing, Zero-Waste Production Models, and Resource-Efficient Systems. This study contributes not only methodologically, but also strategically, by offering a path toward more sustainable, intelligent, and responsive production ecosystems, in line with ICE2025's focus on leveraging AI to promote competitiveness and sustainable innovation.

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